



## ASSET LOADING · SCENE SETUP

Choose a blue sky backdrop and delete the default cat



### STEPS TO FOLLOW

1. Open Scratch → right-click cat → Delete
2. Choose a Backdrop → search "Blue Sky"
3. Click to add → stage turns sky blue!

#### TIP

Any bright sky backdrop works. "Blue Sky 2" has clouds already painted in — saves you one step!

#### DID YOU KNOW?

Loading assets and configuring the scene before the game loop starts is called scene setup or initialisation — every game engine has this explicit phase.



## OBJECT PROPERTIES • POSITIONING

**Add a parrot or bird sprite, size it to 50, and position it on the left of the stage**



### SCRATCH BLOCKS — ON THE BIRD SPRITE

when  clicked

go to x: (-180) y: (0)

#### TIP

Placing the bird at x=-180 gives it room to navigate. Pipes will come from the right side.

#### DID YOU KNOW?

Every game object starts with an initialization block that sets position, size, and velocity to known values before the game loop begins.



## VARIABLES · PHYSICS SIMULATION

Create a "y velocity" variable; every frame subtract 1 (gravity) and apply it to y position



### SCRATCH BLOCKS — ON THE BIRD SPRITE

when  clicked

go to x: (-180) y: (0)

set [y velocity] to (0)

forever

change [y velocity] by (-1)

change y by (y velocity)

if <(y position) < (-170)> then

set y to (-170)

set [y velocity] to (0)

#### TIP

Try "change [y velocity] by (-2)" for stronger gravity — the game gets much harder immediately!

#### DID YOU KNOW?

Velocity accumulation under constant acceleration is Newtonian mechanics —  $F=ma$  in code. Every physics engine from Box2D to Unreal uses this exact loop every frame.



## EVENT HANDLING · INPUT POLLING

Press SPACE to set y velocity to +8 — fighting gravity with a flap impulse



## SCRATCH BLOCKS — SEPARATE SCRIPT ON BIRD

– *separate script on BIRD* →

when [space] key pressed

set [y velocity] to (8)

### TIP

Try values like 6, 10, or 15 to find the feel you want. Higher = easier to stay aloft but harder to aim through gaps.

### DID YOU KNOW?

"when [space] key pressed" is a hardware interrupt wrapper — the operating system tells Scratch the instant the key is pressed, with no polling delay.





## OBJECT TEMPLATES · HIDING

Draw a green pipe sprite in the Paint Editor and hide it on flag click



### SCRATCH BLOCKS — ON THE PIPE SPRITE

— *PIPE*: when  clicked —

hide

#### TIP

Make the pipe width about 28–32 pixels. Too wide = impossible game. Too thin = too easy.

#### DID YOU KNOW?

The Pipe sprite is a class. Its clones are instances. OOP in action — one template, infinite objects.



REPEAT-UNTIL · DESTRUCTION

A pipe clone scrolls from right to left and deletes itself when  
offscreen



### SCRATCH BLOCKS — ON THE PIPE SPRITE

— PIPE: when I start as a clone —

show

repeat until  $\langle (x \text{ position}) < (-240) \rangle$

change x by  $(-4)$

delete this clone

#### TIP

Increase the scroll speed to -6 or -8

for a harder game. This one number

controls overall game difficulty!

#### DID YOU KNOW?

Parallax scrolling, endless runners,

and side-scrollers all use this pattern:

spawn off-screen right → scroll left →

destroy off-screen left → repeat.



## RANDOM NUMBERS • PROCEDURAL GENERATION

Make the master Pipe spawn clones at random heights in a forever loop



### SCRATCH BLOCKS — ON THE PIPE SPRITE

— PIPE master: when clicked —

hide

forever

go to x: (240) y: (pick random (-60) to (60))

create clone of [myself]

wait (2.5) seconds

#### TIP

Lower the wait to 1.5 secs for a rapid-fire challenge. Wider pick-random range creates more extreme height differences.

#### DID YOU KNOW?

Procedural generation — Minecraft biomes, Spelunky levels, No Man's Sky planets — is all "pick random" at different scales.



**COLLISION DETECTION · GAME OVER**

**Detect when the Bird touches a Pipe — stop the game immediately**



### SCRATCH BLOCKS — INSIDE THE PIPE SCROLL LOOP

*(inside the repeat-until scroll loop)*

change x by (-4)

if <touching [Bird]?> then

stop [all]



#### TIP

Before "stop [all]", add "say [Game Over! 🌟] for (1) secs" on the Bird sprite for a dramatic ending.



#### DID YOU KNOW?

"touching [Bird]?" checks pixel-by-pixel overlap 30 times per second — the same AABB collision test behind every hitbox, trigger zone, and collision mesh in game development.





## VARIABLES · SCORE STATE

**Add a Score variable that increments each time the bird passes a pipe**



### SCRATCH BLOCKS — INSIDE THE PIPE CLONE SCROLL LOOP

*(inside the pipe clone scroll loop)*

```
if <(x position) < (-180)> then
```

```
  change [Score] by (1)
```

```
  wait (0.1) seconds
```

```
if <touching [Bird]?> then
```

```
  stop [all]
```

#### TIP

Share your game on Scratch — click "Share" at the top and challenge friends to beat your high score! 🏆

#### DID YOU KNOW?

Threshold-crossing detection ("x < -180") is used in games, finance (stop-loss triggers), IoT (sensor alarms), and analytics (funnel drop-off).